Seismic Vulnerability Assessment of Non-Structural Elements Inside an Inflatable Lunar Habitat

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ABSTRACT

The spirit of exploration and pursuit of knowledge are once again driving humanity to seek a human presence on extraterrestrial bodies. While numerous challenges must be overcome to achieve this goal, the existence of moonquakes has often been overlooked. The Apollo Passive Seismic Experiment (APSE) collected 8 years' worth of data, offering the scientific community insights into the Moon's internal mechanisms. Using NASA's Moon to Mars Architecture requirements for surface habitation, a procedure for vulnerability assessment is developed herein for essential non-structural elements (NSE) inside an inflatable habitat. The NSE element used for this study is the environmental control and life support system (ECLSS). The aim of this study is to propose an approach to assess the vulnerability of the NSE that supports essential equipment under a paucity of information related to the seismic hazard. We also emphasize herein that the launch dynamic environment behaves differently, yielding a need to equip a future habitat with seismic mitigation measures. The findings in this work suggest that due to epistemic uncertainty, significant disturbances may result in long-lasting vibrations that need to be considered as they may require effective mitigation measures.

INTRODUCTION

In recent years, global interest in Lunar exploration has surged, mirroring the enthusiasm of the Apollo era. While Lunar surface missions flourished from the 1950s to the 1970s, the following decades witnessed a hiatus in Lunar exploration. However, the landscape has shifted, with NASA and private entities now heavily investing in returning to the Moon, making it a short-term goal within their broader mission to reach Mars. Under NASA's Artemis program, plans are underway for a short-term surface mission by the end of this decade.

To support these ambitions, it is crucial to explore surface habitat concepts for Lunar missions, with a particular focus on inflatable habitats. Inflatable habitats offer an advantage by significantly increasing payload volume upon deployment on the Lunar surface. The concept of space habitation in inflatable structures has been in existence for decades, but its practical application was first tested with the Bigelow Expandable Activity Module (BEAM) module on the ISS. The BEAM module occupied 3.6 m³ on the launch vehicle and expanded to an operational volume of 16 m³ (Valle et al. 2019; Valle and Wells 2018). Currently, Sierra Space is in the process of developing their Large Integrated Flexible Environment (LIFE) orbital

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habitat, positioned to serve as the next-generation space station after the retirement of the ISS, boasting a claimed operational volume of 270 m³ (Morgan et al. n.d.). NASA, in its Moon-to-Mars Architecture definition document and various other NASA sources, has used the example of a surface habitat consisting of an inflatable torus on top of a lander (NASA 2023b).

Despite these developments, the impact of seismic motions on Lunar habitats has not received attention. Lunar seismic research, stemming from Apollo missions, has uncovered that the Moon experiences seismic activity, being the most critical events for Lunar infrastructure the shallow moonquakes. Dr. Ceri Nunn from NASA's Jet Propulsion Laboratory in California, the lead scientist in the project SeisMo recently re-analyzed the data from the five seismometers left behind by the Apollo astronauts. Based on the observations from the study Dr. Nunn indicated that possibly the upper layer of the Moon’s surface is likely highly fractured, and up to 100 km thick, both of which disturb the movement of seismic waves across the surface. Currently there are no active seismometers on the Moon. The limited data hinders precise vulnerability assessment for Lunar habitats in the face of these shallow moonquakes, and its importance has been a low priority due to the claim that the habitat and its content are designed to survive the rigors of the launch environment vibrations. From the engineering perspective, Wootton and Jablonski 2021, Mottaghi and Benaroya 2015, have concluded on the need to evaluate how moonquakes can affect future Lunar infrastructure. This raises a critical question, given that inflatable Lunar habitat designed for short-to-long-term missions is engineered for launch vibration conditions, will shallow moonquakes pose a significant risk to its structural integrity? If so, how should this vulnerability be assessed to determine the potential impact on the habitat's contents?

This paper aims to address these concerns, comparing the performance of Class I Lunar habitats under launch vibrations and shallow moonquakes. We present a case study, analyzing habitat performance under these two scenarios and outlining an approach for assessing vulnerability concerning the mission duration. The results offer valuable insights for Lunar surface missions.

CASE STUDY

In this study, the Class I habitat chosen for analysis comprises a torus-shaped inflatable structure mounted atop a lander housing the environmental control and life support system (ECLSS) universal pallet design from Collins Aerospace (O’Neill et al. 2019). The torus stands at a height of 6 meters and encompasses two floors, supported by a cantilever hydraulic jack system. The core of the torus consists of solid aluminum vertical members, leaving open spaces for equipment installation. The connection between the torus and the lander is assumed to be facilitated by an aluminum pedestal. For reference, the lander is assumed to have dimensions similar to those of the descent stage of the Lunar module (LM). Notably, the Lunar habitat exhibits distinct dynamic configurations for both launch conditions and when fully deployed on the Lunar surface (see Figure 1). Within the habitat, the ECLSS universal pallet studied is situated on the second floor.

The ECLSS universal pallet is composed of a frame featuring four aluminum elements, incorporating lateral restraints both at the supports and mid-height points. The dimensions of the pallet are assumed as 0.60mx0.60mx1.80m [25”x25”x70”] according to the suggestions from (O’Neill et al. 2019). It is assumed that the pallet will be supported at both its top and bottom

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via pin connections (see Figure 2a-b). The universal pallet is assumed to be positioned within the aluminum core of the torus, as this space remains unchanged between deployment conditions.

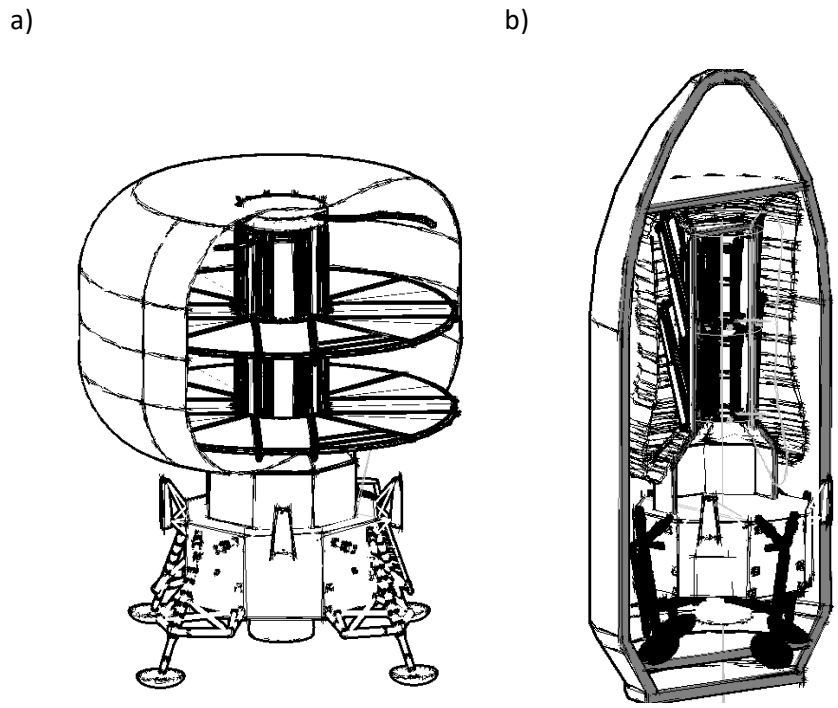

Figure 1. Case of study conceptual views: a) 3D view deployed. b) 3D view undeployed.

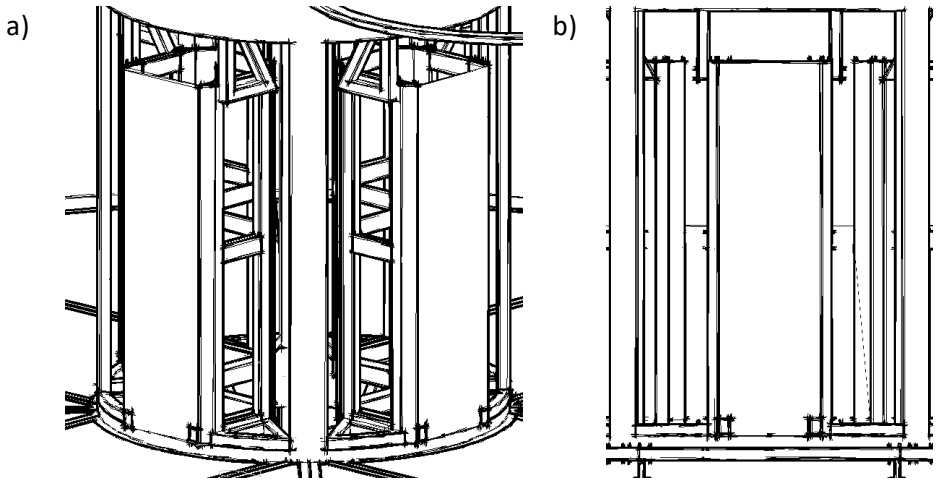

Figure 2. Internal non-structural elements: a) 3-D view and b) front view representation of the ECLSS universal pallet.

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A time-stepping technique with modal simplification using Newmark's method is employed to solve the fundamental dynamic equation of motion. The damping matrix is constructed with the assumption of classical damping at 5% of critical damping. Consequently, it is important to thoroughly examine the dynamic loads imposed by the launch environment on the universal pallet. This step is crucial for establishing a reference point for assessing the integrity of the state-of-the-art structural design of the habitat towards the seismic motions that may experience on the Lunar surface after deployment.

LAUNCH DYNAMIC ENVIRONMENT

In the context of rocket launches, payloads face a range of dynamic loads, including those stemming from engine ignition, liftoff release, aerodynamic pressures, and maneuvering forces. These loads originate from various sources, such as acoustic and vibration excitations, and must be carefully assessed to ensure the structural integrity and functionality of the spacecraft and its payloads. All of these loads are encapsulated in a set of test requirements that hinge on the specific characteristics of the payload (NASA, 2023). Among other required vibration tests, the sine sweep test and the random vibration test are most common.

Commercial spacecraft such as the Atlas V and Falcon vehicles provide initial recommendations for the sine vibration test, which can later be refined through the use of couple load analysis (CLA), and offer guidelines for high-frequency random vibration tests (SpaceX, 2021; United Launch Alliance, 2010). To generate the sine sweep time history, specifications for lateral loads from both companies are adhered to, with a duration matching the rate of 4 octaves per minute (NASA, 2017) (see Figure 3). Similarly, the synthesizing of a time history in accordance with the power spectral density (PSD) outlined in NASA's standard minimum workmanship criteria and Falcon vehicle recommendations is used, with a set duration of 2 minutes, following NASA's stipulated requirements (NASA, 2017) (see Figure 4). It is worth noting that accurately representing the dynamic forces experienced during launch is a complex task, and relying solely on these two signals serves as a foundational approach for designing the payload to withstand lateral loads. A Class I habitat situated within the payload fairing of such rockets will need to endure these forces without damage, but it's important to consider that their ultimate goal is an extended stay on the Lunar surface, where seismic motions constitute another source of vertical and horizontal motions. Therefore, it is important to describe these seismic signals in detail.

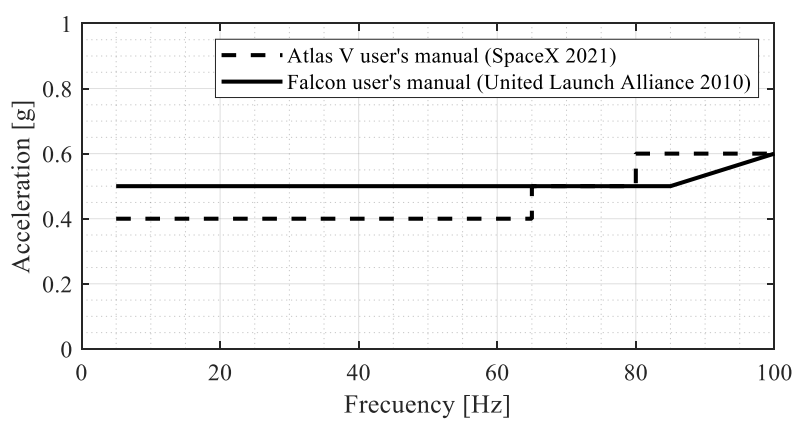

Figure 3. Sine sweep vibration test signal: Users guide specification seed from space vehicle manufacturer.

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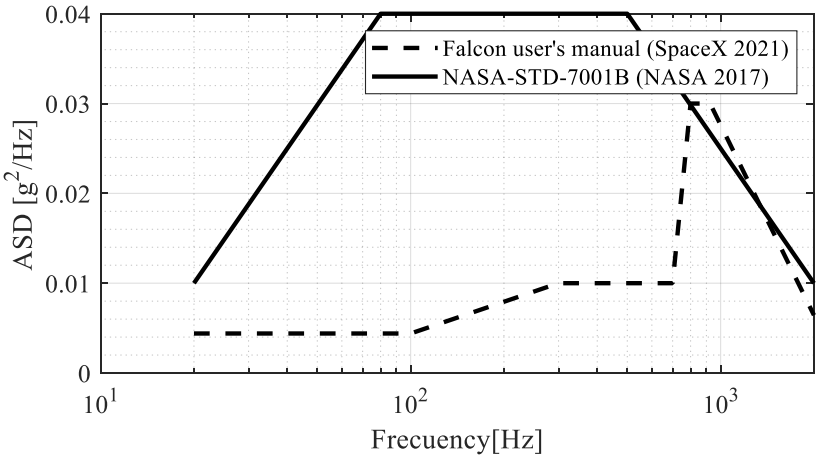

Figure 4. Random vibration test signal: Mass attenuated workmanship vibration spectrum from NASA-STD-7001B.

LUNAR SEISMIC MOTION ENVIRONMENT

The seismic data collected by the Apollo Passive Seismic Experiment (APSE) was initially recorded on analog tape. To ensure accuracy and reliability, we employed the raw seismic data from (Nunn et al., 2022), which was formatted in the Standard for the Exchange of Earthquake Data (SEED). Furthermore, we adopted the data processing and refinement methodology outlined in their work for the purpose of cleaning and analyzing the seismic data. Among the extensive seismic dataset comprising approximately 12,000 events (Nakamura et al. 1982), the seismic motions of particular interest are the 28 shallow moonquakes, also known as High-Frequency Teleseismic events (HFT). These events represent the most energetic seismic occurrences observed during the 8-year campaign.

We initiated the process by selecting seismic motions for our analysis. To achieve this, we followed the state-of-the-art recommendation of using eleven distinct motions to capture various motion characteristics. In Figure 5, is noticeable the comprehensible expectation that the amplitudes experienced from shallow moonquakes by a surface habitat launched from Earth should already be well-suited to withstand these lateral loads, given their survival during launch.

However, it is crucial to highlight that all these events occurred at distances as far as 500 km from the measuring station, with the majority of them being recorded at distances of at least 1000 km from the epicenter (Oberst 1987). Seismic waves traveling through the Lunar crust experience attenuation with distance, meaning that ground acceleration measurements decrease as the distance from the seismic station increases. Unfortunately, there is insufficient seismometer data to confidently establish these relationships. However, some researchers have explored this issue by observing satellite images. They have discovered that in close proximity to the seismic event source, ground acceleration can reach up to $1.6g_L$ (Watters et al. 2019), and for a moment magnitude event of M_w=4, this acceleration can increase to as much as $4g_L$ (Mohanty et al. 2020). Both sources concur that beyond 50 km from the source, the acceleration levels begin to flatten out, approaching zero. Drawing conclusions about seismic motions using the directly measured data would be an oversimplification. Pointing out the need to collect more

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data to properly characterize the loadings associate with this hazard. To accurately predict seismic hazard, it is necessary to investigate variations in motion amplitude with distance and the occurrence patterns of these events. This comprehensive analysis aligns with the principles of earthquake engineering, particularly the probabilistic seismic hazard analysis (PSHA).

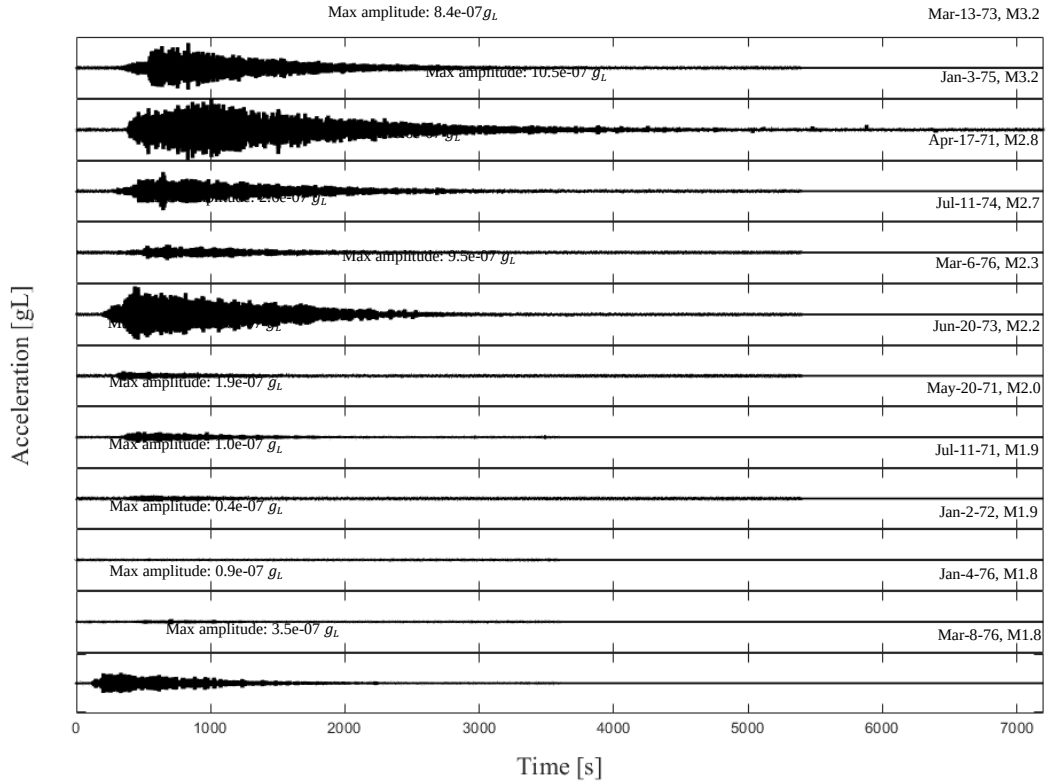

Figure 5. Shallow seismic events used in the analysis.

ASSESSMENT OF LUNAR SEISMIC VULNERABILITY

The engineering seismic risk analysis requires the hazard characterization and the identification of vulnerabilities. A widely used and accepted method to characterize the hazard is detailed in (Cornell, 1968). This approach has undergone significant evolution over the decades, with contributions from various authors culminating in what is now known as Probabilistic Seismic Hazard Analysis (PSHA). To this day, PSHA relies on two main inputs: Ground Motion Prediction Equations (GMPEs) and site-specific magnitude-recurrence data. The accuracy and reliability of the method are heavily contingent upon the quality of these inputs. Following its global recognition, ongoing efforts have focused on gathering sufficient data for the site of interest to establish robust magnitude-recurrence estimations and achieving an accurate representation of the GMPE for the site through geophysical models of the specific seismic setting. The method to this day is represented with the following equation:

$$P(Y > y) \cong \sum v_i \iint P[Y > y|M, R]\, f_{M,R}(m, r)\, dm\, dr$$

Where Y and y are earthquake ground motions, M and m are magnitudes, R and r are distance, v_i is the rate of earthquakes for source i, and the summation is over all sources.

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To construct an accurate seismic model, it is essential to acquire more data in close proximity to the targeted site. For instance, consider the state of California in the United States, which spans an area of approximately 5000 km^2, similar in scale to the Lunar South Pole. The state's seismic instrumentation has evolved significantly over the years. In the 1930s, there were around 50 instruments in place, and by the 1970s, that number had grown to 500. Today, the state boasts more than 1000 instruments. In the case of Ground Motion Prediction Equations (GMPEs) for California, the wealth of data collected over several decades has enabled substantial advancements. In the 1980s, they had only a rudimentary model of the internal seismic mechanism, primarily based on 16 known fault sources. By the 2000s, this number had increased to 200, and in the most recent model, they've incorporated 350 faults and the interactions between them. This progression highlights the importance of accumulating data to develop accurate seismic risk models and underscores that understanding the internal mechanisms requires significant time and research efforts.

On the Moon, the APSE recorded a total of 28 shallow moonquakes over an 8-year period (see Figure 6). This limited frequency-occurrence relationship serves to date as the foundation for our understanding. On Earth, a comprehensive interpolation process is typically employed, leveraging a multitude of data points from seismometers located in close proximity to one another. However, on the Moon, such an approach is not feasible due to the dearth of instrumentation to gather such information. There, seismometers were positioned as far as 1000 km apart, rendering it impossible to draw conclusive results for a specific site. The conclusion from Dr. Nunn's study indicating the possible highly fractured condition of the first 100 km of the Lunar crust introduces further uncertainty to any seismic hazard considerations (O'Callaghan 2020). Similarly, the construction of GMPEs on Earth relies on data collected from seismometers recording varying peak ground accelerations, which can be correlated with their distance from the seismic source to formulate predictive curves. As noted, this approach is also hindered by the limited data available thus far.

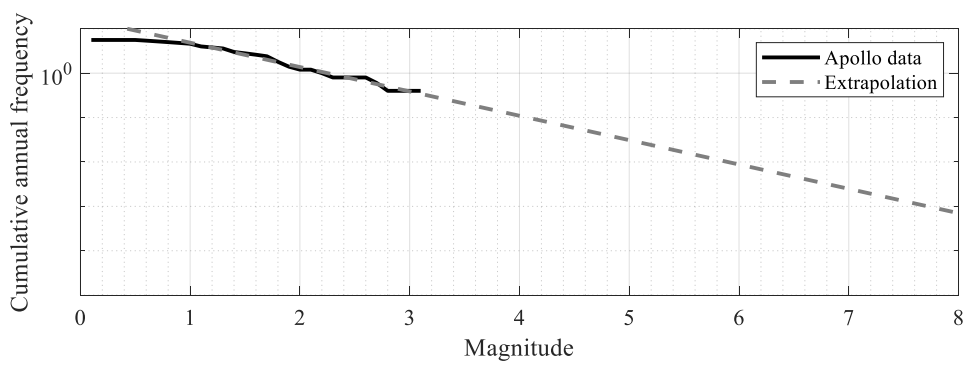

Figure 6. Frequency-occurrence relationship from the APSE.

Probabilistic Seismic Hazard Analysis (PSHA) produces a uniform hazard spectrum (UHS) that relates the return period to a single-degree oscillator, representing the response of a building, or as in this case the lunar habitat. This spectrum is based on a predetermined damping coefficient and provides an envelope for both the frequency content and maximum amplitude of expected seismic motions at specific sites (Baker et al. 2021; Cornell 1968). This tool enables engineers to design structures for a selected probability of occurrence, within the limits of the method. This concept was initially depicted for the Lunar conditions in the work of Ruiz et al.

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(2022) for a specific site, utilizing Earth-based formulations for the method's unknowns. They proposed a design spectrum near the landing site of Apollo 17 for various levels of probability of occurrence. However, this method is intended to process a vast amount of seismic motion data and simplify it into an engineer-applicable design tool. Subsequently, engineers can select motions near a specific site and scale their frequency content and amplitude to the desired range based on the fundamental frequency of the structure. On the Moon, this approach is still in its early stages due to the lack of data.

Considering the current infancy of applying PSHA on the Moon, a process to assess the vulnerability of a class I inflatable Lunar habitat is presented (see Figure 7). The proposed process for assessment of seismic vulnerability is aimed at structural engineers with the goal of establishing the impact of a strong motion would represent for critical non-structural habitat contents. The step "design ground acceleration" refers to the peak ground acceleration of each of the 28 recorded events, scaled only in amplitude, without altering the frequency content. However, this scaling introduces uncertainty into the procedure, as the frequency content of the scaled motion may differ. Consequently, the proposed process entails iteratively assuming an amplitude scale for these 28 events until it aligns with the design requirements for the launch environment. This iterative approach allows engineers to, based on their judgment, evaluate the risk of not implementing mitigation measures for the habitat. It highlights the needed amplitude scale, while maintaining the frequency content, to overcome the known lateral design of the launch environment.

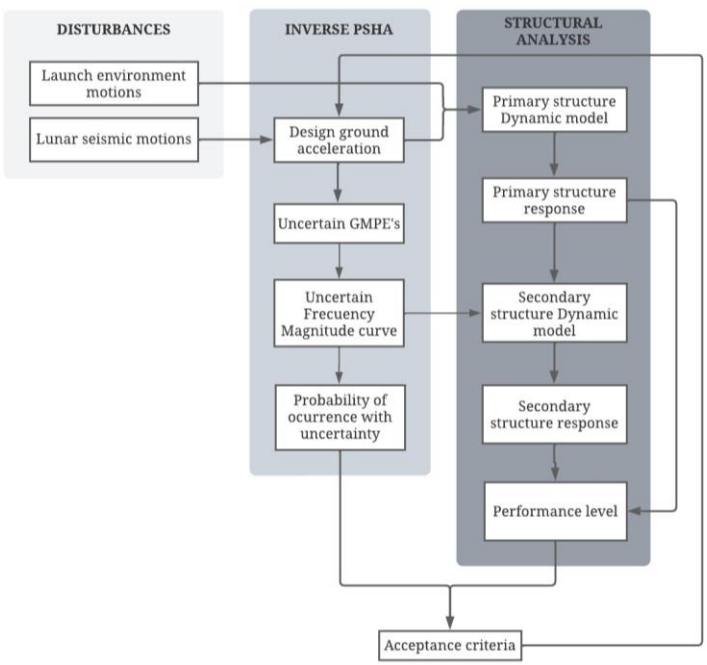

Figure 7. Process for Lunar seismic vulnerability assessment of a case of study.

NUMERICAL SIMULATION

Implementing the procedure on the case study, as defined in the previous section, and considering all aspects of the analysis, reveals that seismic motion produces distinct responses

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in terms of amplitude, frequency content, and duration for the Class I Lunar habitat. Additionally, the seismic motion introduce a risk into the habitat's design associated with the mission duration due to the levels of uncertainty inherent in the engineering process. Following the steps identified in Figure 7, the disturbances are determined based on the definitions provided in earlier sections. An inverse PSHA is conducted, followed by a structural analysis using these ground accelerations. Finally, an arbitrary acceptance criteria is established to qualify the responses.

First, the response of the primary structure is evaluated using a recorded event from the Lunar seismic environment, with a scaled value of $1.0g_L$ arbitrarily selected as the maximum expected peak ground acceleration for the habitat. It is worth noting that various authors suggest that near the source, this parameter could reach far mor than $1.0g_L$ (Mohanty et al. 2020; Watters et al. 2019). The deployed structure, shown in Figure 8, exhibits a maximum displacement of 2.4cm, while the undeployed structure, within the payload fairing, experienced an estimated maximum displacement of 0.6cm. Furthermore, the displacement response time histories reveal that this response due to the Lunar seismic environment lasts for over an hour, whereas the launch environment's effect only persists for a few minutes.

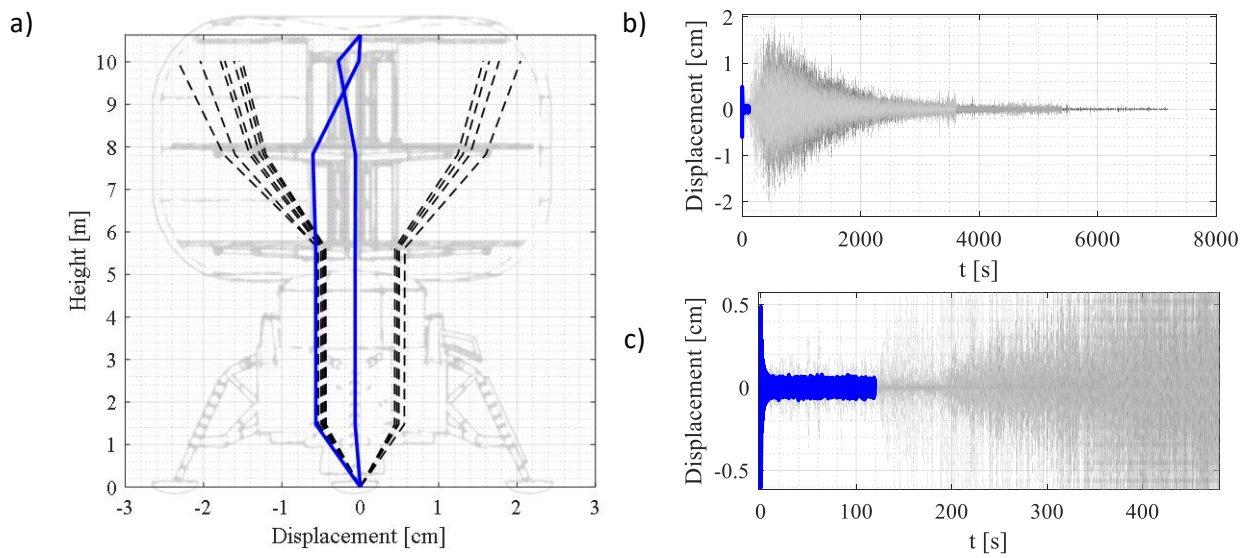

Figure 8. Deployed class I inflatable Lunar habitat response - (a) Deformed shape, (b) Time history of maximum displacement, and (c) zoom-in on the early seconds of the previous time history. Solid lines represent the launch environment responses, while dashed lines correspond to the eleven Lunar seismic environment responses.

Recognizing these variations in response, it becomes evident that there must be a scale factor for the Lunar seismic environment that can produce results that are similar to the launch environment, as the latter is already incorporated into the design guidelines and requirements. Following an iterative process, a scale factor of $3.0g_L$ is identified as the ground acceleration that closely approximated the deformation responses of the structures. With the responses obtained for the primary structure under consideration of both scale parameters, $1.0g_L$ and $3.0g_L$, the accelerations at the supports of the non-structural elements were transferred as ground accelerations for the latter. Subsequently, the responses of the ECLSS under both scaling

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parameters are presented. In the case of $1.0g_L$, the deformed shape indicated a maximum displacement of 0.6cm for the deployed state and 0.1cm for the ECLSS inside the undeployed habitat (see Figure 9). Transitioning to the $3.0g_L$ case, the response shape remained similar, with displacements of approximately 0.6cm for both scenarios.

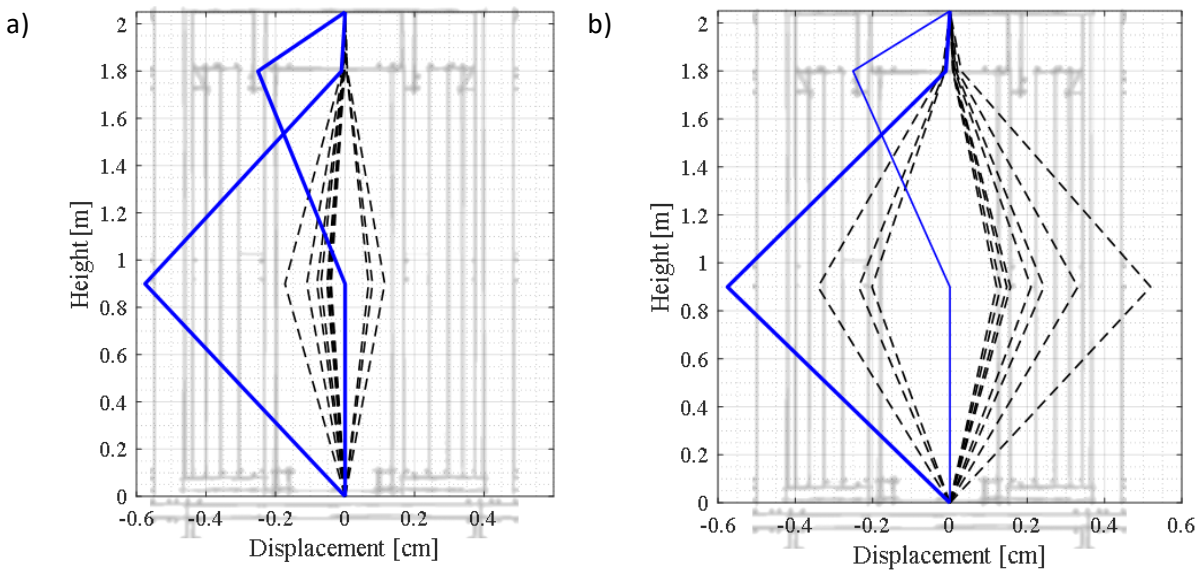

Figure 9. ECLLS deformed shapes - a) Scaled Lunar environment to $1.0g_L$, b) Scaled Lunar environment to $3.0g_L$. Solid lines represent the launch environment responses, while dashed lines correspond to the eleven Lunar seismic environment responses.

DISCUSSION

The dynamic conditions during launch and within the Lunar surface habitat exhibit significant variations in terms of both amplitude and frequency content of dynamic forces. Employing the proposed approach allows for a comprehensive assessment of the potential impact of shallow seismic motions on the habitat's contents concerning different mission durations.

The Lunar seismic environment primarily operates at low frequencies compared to the launch dynamic environment, less than 10Hz, which can resonate with certain Lunar surface habitat designs, similar to the case study utilized in this work. Additionally, the prolonged duration of Lunar seismic events could have further implications for material integrity, particularly in relation to fracture and fatigue. Further studies in this regard could be warranted.

It is evident that, depending on the dynamics of the primary structure and the non-structural elements, they may resonate with the frequency content of the lunar seismic environment. These frequencies are not accounted for in the launch vibration design. As depicted in the case study, the primary structure exhibited a greater transverse response for $1.0g_L$ than anticipated during launch. However, the ECLSS, with a higher fundamental frequency, had greater lateral response from the launch-induced vibrations. This disparity shows how susceptible the response is to both the primary structure and non-structural elements design. Therefore, there is a need for future surface settlements, especially for long-term use, to consider the effects of the Lunar seismic environment, as more seismic data is collected. The approach proposed here provides a tool for

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engineers to adopt. ECLSS and other non-structural but fundamental systems must maintain functionality during and after any seismic motions.

The results using the process could mean in qualitative terms that the Lunar seismic environment produces a different requirement for lateral loads in the design of the habitat, and in quantitative terms, that the uncertainty range of the results is too high due to a lack of information on the site. This means that ignoring the risk of the Lunar seismic environment and only considering the launch environment could result in undesired outcomes for the habitat and contents for extended periods of performance.

A primary design trade made for space structures is to function effectively for mission objectives while optimizing material usage. This design philosophy conflicts with seismic design principles, which aim to accommodate the maximum expected motion. This approach is fundamentally problematic in a seismic context, where the optimization of a non-deterministic force inherently involves a probabilistic dimension. This work highlights the need to refine probabilistic methods in Lunar seismic assessment to better understand their implications for overall structural design. Future work will need to address these issues.

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